

S03 SOUND BARRIERS | O (MAX : 3 PT)

Intent : Increase the level of sound isolation and speech privacy between enclosed spaces.

Summary : This WELL feature requires that walls and doors meet a minimum degree of acoustical separation to provide adequate sound isolation and improve speech privacy.

Issue : Sound that transmits from one room to another through walls or doors can be distracting or annoying and also disturb sleep.¹⁻⁵ There is evidence that links noise annoyance in multi-story housing to poor mental health and perceived stress in residents.^{6,7} Speech privacy is also reduced when background sound in receiving rooms is lower as part of the room's intent (e.g., bedroom, restorative space, classroom).^{1,8} For walls, lightweight construction, glass and demountable partitions are typical in modern design and offer minimum acoustic separation and speech privacy, especially when wall construction is not uniform.⁹⁻¹¹

Solutions : Sound transmits through walls directly and around the construction by what is known as flanking. Walls with high sound transmission class ratings (STC) will provide sound isolation, only when the wall is constructed to reduce flanking at points where the wall connects to other building elements. Windows, glass fins or other penetrations diminish performance and should be used sparingly. Doors also reduce performance of walls but can be fitted with gaskets and seals to reduce sound transmission, when the door is closed. Rooms that require high speech privacy can use sound masking systems, in addition to high-performing walls to increase privacy.⁸

PART 1 DESIGN FOR SOUND ISOLATION AT WALLS AND DOORS (MAX : 1 PT)

For All Spaces:

The following requirements are met:

- Interior walls meet the following sound transmission class (STC) or weighted sound reduction (Rw) values. If an interior wall meets multiple categories listed, use the highest (i.e., more stringent) STC/Rw value listed.

Interior Wall Type	Minimum STC or Rw	OR	Minimum STC or Rw for Projects That Achieve Feature S06.1
Between loud zones and regularly occupied spaces.	60	OR	60
Between areas for conferencing, learning or sleeping and other regularly occupied spaces.	55	OR	50
Between adjacent quiet zones.	50	OR	45
Between rooms for concentration and other regularly occupied spaces.	45	OR	40
Between circulation zones and regularly occupied spaces.	40	OR	35

- Doors that connect two occupiable rooms and doors to mechanical equipment rooms have a non-hollow core, minimum STC/Rw of 30 and seals at the head, jamb and base.

PART 2 ACHIEVE SOUND ISOLATION AT WALLS (MAX : 2 PT)

For All Spaces:

Option 1: Noise isolation class

For walls that separate regularly occupied spaces the following requirements are met:

- The project meets the following minimum Noise Isolation Class (NIC) or Weighted Difference Level (Dw) for each wall type, as applicable. If an interior wall meets multiple categories listed, use the highest NIC/Dw value listed.

Interior Wall Type	Minimum NIC or Dw	OR	Minimum NIC or Dw for Projects That Achieve Feature S06.1
Between loud zones and regularly occupied spaces.	55	OR	N/A
Between areas for conferencing, learning or sleep and other regularly occupied spaces.	50	OR	45
Between adjacent quiet zones.	45	OR	40
Between rooms for concentration and other regularly occupied spaces.	40	OR	35
Between circulation zones and regularly occupied spaces.	35	OR	30

OR

Option 2: Speech privacy

For walls that separate regularly occupied spaces the following requirements are met:

- The sum of the measured Noise Isolation Class (NIC) or Weighted Difference Level (Dw) combined with the Noise Criteria Rating (NC) or A-weighted Sound Pressure Level (LAeq) within a room achieves the following minimum values, as applicable. If an interior wall meets multiple categories listed, use the highest value listed.

Source Room	Receiver Room	Minimum NIC + NC or Dw + LAeq
Enclosed loud zones	Open areas for concentration	80
	Circulation zones	
	All other occupiable areas	85
Enclosed areas for conferencing	Open areas for concentration	75
	Circulation zones	
	Enclosed quiet zones	80
Enclosed areas for learning	Enclosed areas for conferencing	85
Enclosed areas for sleeping	Enclosed areas for learning	
	Enclosed areas for sleeping	
Enclosed quiet zones	Open areas for concentration	70
	Circulation zones	
	Enclosed quiet zones	75
	Enclosed areas for conferencing	
	Enclosed areas for learning	80
	Enclosed areas for sleeping	

Note :

Refer to the Performance Verification Guidebook for information on sensor/testing requirements, required testing duration and compliance calculations.

REFERENCES

1. Hongisto V, Oliva D, Keränen J. Subjective and objective rating of airborne sound insulation - living sounds. *Acta Acust united with Acust.* 2014;100(5):848-863. doi:10.3813/AAA.918765
2. Park HK, Bradley JS, Grover BN. Rating Airborne Sound Insulation in Terms of the Annoyance and Loudness of Transmitted Speech and Music Sounds.; 2008.
3. Xie H, Kang J, Mills GH. Clinical review: The impact of noise on patients' sleep and the effectiveness of noise reduction strategies in intensive care units. *Crit Care.* 2009;13(2):208. doi:10.1186/cc7154
4. Strauch C, Brandt S, Edwards-Beckett J. Implementation of a quiet hour: effect on noise levels and infant sleep states. *Neonatal Netw.* 1993;12(2):31-35.
5. Parthasarathy S, Tobin MJ. Sleep in the intensive care unit. *Intensive Care Med.* 2004;30:197-206.
6. Jensen HAR, Rasmussen B, Ekholm O. Neighbour noise annoyance is associated with various mental and physical health symptoms: results from a nationwide study among individuals living in multi-storey housing. *BMC Public Health.* 2019;19(1):1508. doi:10.1186/s12889-019-7893-8
7. Jeon JY, Ryu JK, Lee PJ. A quantification model of overall dissatisfaction with indoor noise environment in residential buildings. *Appl Acoust.* 2010;71(10):914-921. doi:10.1016/j.apacoust.2010.06.001
8. Hongisto V, Varjo J, Leppämäki H, Oliva D, Hyönä J. Work performance in private office rooms: The effects of sound insulation and sound masking. *Build Environ.* 2016;104(1):263-274. doi:10.1016/j.buildenv.2016.04.022
9. Trevathan JW, Pearse JR. The effect of workmanship on the transmission of airborne sound through light framed walls. *Appl Acoust.* 2008;69(2):127-131.
10. Craik RJM, Steel JA. The effect of workmanship on sound transmission through buildings: Part 1 - Airborne sound. *Appl Acoust.* 1989;27(1):57-63.
11. Craik RJM. The consequences of lightweight buildings on sound transmission. *Build Serv Eng Res Technol.* 1998;19(1):B1-B4.